FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2023

TEST DATE: 02-04-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

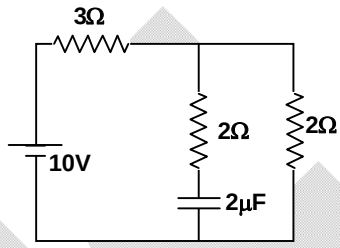
PART – A

SECTION – A

(Single Choice Answer Type)

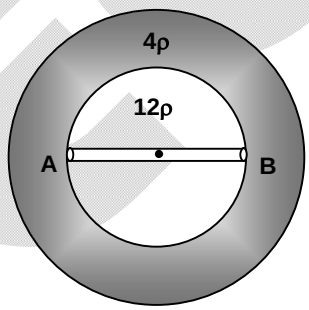
This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. In the circuit shown here, what is the charge on the capacitor in steady state?



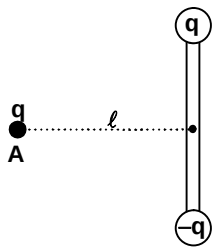
(A) 2 μC (B) 4 μC
(C) 6 μC (D) 8 μC

2. A planet of core density 12ρ and outer crust density 4ρ has a small tunnel in core. A small particle is released from end A then time required to reach end B, is



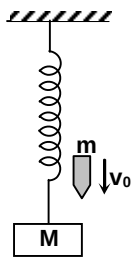
(A) $\sqrt{\frac{\pi}{\rho G}}$ (B) $\frac{1}{2}\sqrt{\frac{\pi}{\rho G}}$
(C) $\frac{1}{4}\sqrt{\frac{\pi}{\rho G}}$ (D) $\frac{1}{6}\sqrt{\frac{\pi}{\rho G}}$

3. Figure shows a point charge 'q' fixed at point A. Two identical ball of mass 'm' having charge q and –q are attached to the end of a light rod of length ℓ which is free to rotate about its fixed centre. If the system is released from situation shown in the figure, what is the angular velocity of rod when it turns through 90°?



(A) $\sqrt{\frac{4q^2}{3\pi\varepsilon_0 m\ell^3}}$ (B) $\sqrt{\frac{q^2}{3\pi\varepsilon_0 m\ell^3}}$
(C) $\sqrt{\frac{2q^2}{3\pi\varepsilon_0 m\ell^3}}$ (D) $\sqrt{\frac{3q^2}{4\pi\varepsilon_0 m\ell^3}}$

4. For the given system block is initially at rest as shown in the figure, a bullet strikes the block inelastically and sticks to it. Neglecting the thickness of the block, find the amplitude of the oscillation of the system. (mass of block = M and mass of bullet = m)



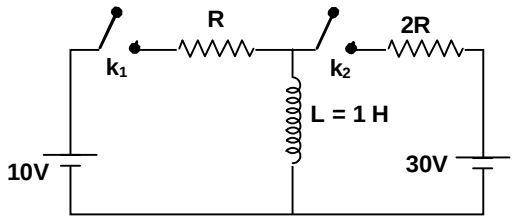
(A) $\frac{m}{k}\sqrt{\frac{v_0^2}{M+m}+g^2}$ (B) $\frac{2m}{k}\sqrt{\frac{v_0^2}{M+m}+g^2}$
(C) $\frac{m}{k}\sqrt{\frac{v_0^2 k}{M+m}+g^2}$ (D) $\frac{2m}{k}\sqrt{\frac{v_0^2 k}{M+m}+g^2}$

5. A student has 4 set of tuning forks, one with lowest frequency vibrating at 552 Hz. By using any two forks at time, the beat frequencies heard are 1, 2, 3, 5, 7, 8. The possible frequencies of other three forks are

(A) 551 Hz, 554 Hz and 560 Hz (B) 553 Hz, 554 Hz and 560 Hz
(C) 553 Hz, 556 Hz and 558 Hz (D) 553 Hz, 555 Hz and 560 Hz

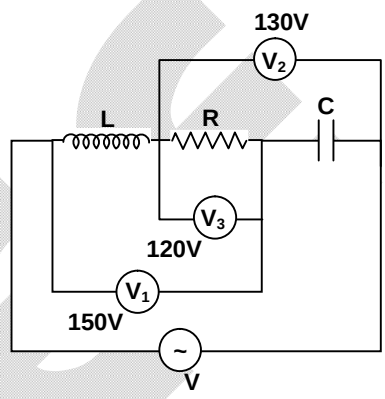
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6. In the given circuit, key k_1 remains connected for a long time and k_2 was open. Now the k_2 is also closed. Find the magnitude of rate of change of current in inductor just after key k_2 is closed. (take L = 1H)



(A) 10 A/s
(B) 15 A/s
(C) 25 A/s
(D) 30 A/s

7. In the circuit shown in figure, the reading of voltmeters, V_1, V_2, V_3 are 150 V, 130 V and 120 V respectively. Find the applied AC voltage (V) and power factor (cos ϕ) of the circuit.



(A) $30\sqrt{7}$ V, $\frac{3}{\sqrt{8}}$

(B) $30\sqrt{10}$ V, $\frac{3}{\sqrt{7}}$

(C) $40\sqrt{10}$ V, $\frac{3}{\sqrt{10}}$

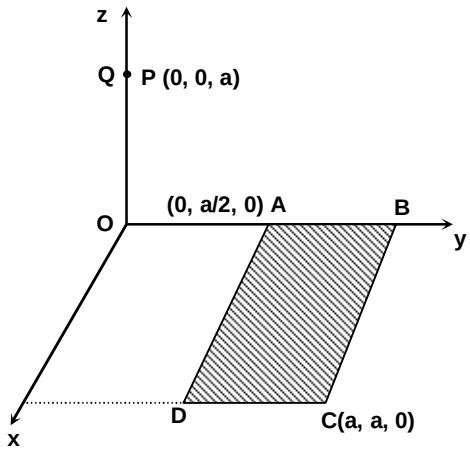
(D) $50\sqrt{10}$ V, $\frac{7}{\sqrt{10}}$

8. A closed organ pipe of length ℓ is sounded together with another closed organ pipe of length $\ell + x(x << \ell)$. Both vibrate in their fundamental modes. If the speed of sound is V, then the beat frequency heard is

(A) $\frac{vx}{2\ell^2}$ (B) $\frac{vx}{4\ell^2}$

(C) $\frac{vx^2}{2\ell}$ (D) $\frac{vx^2}{4\ell}$

9. A charge Q is placed at P(0, 0, a), as shown in the figure. ABCD is a rectangle where coordinate of A and C are $\left(0, \frac{a}{2}, 0\right)$ and (a, a, 0) respectively. What is the flux through ABCD?



(A) $\frac{Q}{16\varepsilon_0}$

(B) $\frac{Q}{24\varepsilon_0}$

(C) $\frac{Q}{48\varepsilon_0}$

(D) none of these

10. The magnetic field of a plane electromagnetic wave is given as,
$\vec{B} = 2\times10^{-8}\sin\left[200\pi(y+ct)\right]\hat{i}T$
Where c = 3×10^8 m/s is the speed of light. The corresponding electric field is

(A) $\vec{E} = 6\sin\left[200\pi(y+ct)\right]\hat{k}$ V/m (B) $\vec{E} = -1.5\times10^{-6}\sin\left[200\pi(y+ct)\right]\hat{k}$ V/m

(C) $\vec{E} = 2\times10^{-8}\sin\left[200\pi(y+ct)\right]\hat{k}$ V/m (D) $\vec{E} = -6\sin\left[200\pi(y+ct)\right]\hat{k}$ V/m

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11. When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by
$y(t) = y_0 \sin^2 \omega t$, where 'y' is measured from lower end of unstretched spring. Then time period of SHM is

(A) $4\pi\sqrt{\frac{y_0}{g}}$ (B) $2\pi\sqrt{\frac{y_0}{g}}$

(C) $2\pi\sqrt{\frac{2y_0}{g}}$ (D) $2\pi\sqrt{\frac{y_0}{2g}}$

12. Two non-identical blocks placed over each other of different densities σ_1 and σ_2 are immersed in a fluid of density σ. The block of density σ_1 is fully submerged and the block of density σ_2 is partially submerged so that the ratio of their mass is $\frac{1}{2}$ and $\frac{\sigma}{\sigma_1} = 2$, $\frac{\sigma}{\sigma_2} = 0.5$. Find the degree of submergence of the upper block of density σ_2.

(A) 25% submerged (B) 50% submerged
(C) 75% submerged (D) 100% submerged

13. A uniform rod of mass 'm' and length 'ℓ' is rotating with constant angular velocity 'ω' about an axis which passes through its one end perpendicular to the length of the rod. The area of cross section of the rod is A and its young's modulus is Y. What is the strain at the mod point of the rod?

(A) $\frac{m\omega^2\ell}{4AY}$ (B) $\frac{3m\omega^2\ell}{4AY}$

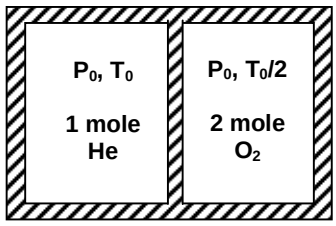
(C) $\frac{m\omega^2\ell}{8AY}$ (D) $\frac{3m\omega^2\ell}{8AY}$

14. A container filled with air pressure P_0 contains a soap bubble of radius R. The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap solution is σ, what is the value of P_0?

(A) $\frac{16\sigma}{R}$ (B) $\frac{25\sigma}{16R}$

(C) $\frac{96\sigma}{R}$ (D) $\frac{125\sigma}{16R}$

15. An adiabatic piston divides a cylindrical container with adiabatic walls into two equal parts. Left part contains 1 mole of helium at temperature T_0 and pressure P_0. The right part contains 2 moles of oxygen at temperature $\frac{T_0}{2}$ and pressure P_0. Now the piston is removed and gases mix homogenously. Find the final pressure.

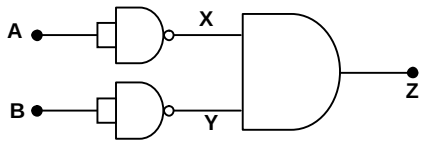


(A) $\frac{4P_0}{13}$ (B) $\frac{12P_0}{13}$

(C) $\frac{2P_0}{13}$ (D) $\frac{12P_0}{23}$

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16. Identify the logic operation carried out by the given circuit.



(A) AND
(B) NAND
(C) OR
(D) NOR

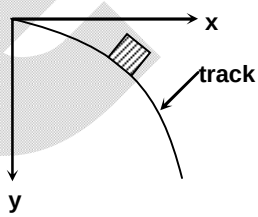
17. A carrier wave with amplitude of 300 V is amplitude modulated by a sinusoidal base band signal of amplitude 200 V. Find the ratio of maximum to minimum amplitude for the amplitude modulated wave

(A) 5 (B) 3
(C) 10 (D) 6

18. Sodium light of wavelength 670 nm and 675 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 1.0 m. What is the separation between the positions of the first maxima of diffraction pattern obtained in the two cases?

(A) 1.5×10^{-5} m (B) 3×10^{-5} m
(C) 4.5×10^{-5} m (D) 6×10^{-5} m

19. A block M slide down on a curved frictional vertical track, starting from rest. The curve obeys the equation $y=\frac{x^2}{2}$. The tangential acceleration of the block is



(A) $\frac{g}{2}$ (B) g
(C) $\frac{gx}{\sqrt{x^2+1}}$ (D) $\frac{gx}{\sqrt{x^2+4}}$

20. A constant external torque τ acts for a very brief period Δt on a rotating system having moment of inertia I. If the system was initially at rest, find the kinetic energy acquired by it.

(A) $\frac{(\tau\Delta t)^2}{2I}$ (B) $\frac{(\tau\Delta t)^2}{4I}$
(C) $\frac{2(\tau\Delta t)^2}{I}$ (D) $\frac{4(\tau\Delta t)^2}{I}$

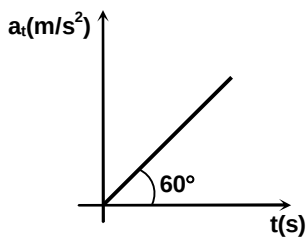
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

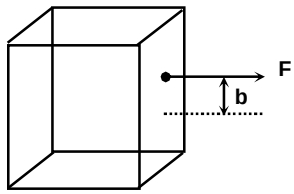
21. The x-coordinate of a particle moving on x-axis is given by
$x=3\sin(100t)+8\cos^2(50t)$, where x is in cm and t is time in seconds.
If the amplitude of the resultant SHM is A in meter, find 100 A

22. Tangential acceleration of a particle starting from rest and moving in a circle of radius 1 m varies with time as shown in the graph. If the time after which total acceleration of particle makes an angle of 30° with radial acceleration in seconds. Find t^3.

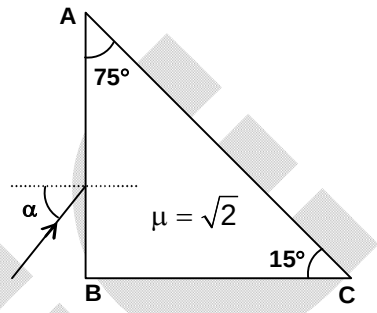


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23. Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see in figure). If coefficient of friction is $\mu = 0.4$, the minimum possible value of $\frac{80b}{a}$ for box not to topple before moving.



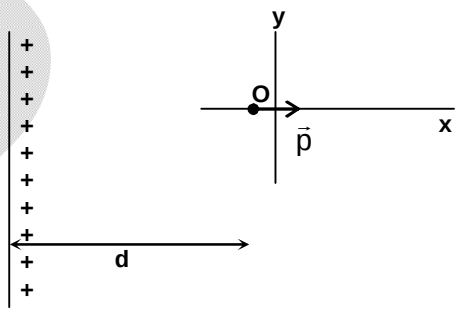
24. In the figure, if ray does not emerge from side AC then maximum value of α is $m\pi$. Find 20 m.



25. An ideal diatomic gas undergoes a process in which density ρ and internal energy U vary as ρU = constant. The ratio of change in internal energy and work done by the gas is x. Find 10x.

26. A stationary Pb^{200} nucleus emits an α-particle. If f is the fraction of total energy liberated in this decay which is accounted for the recoil energy of the daughter nucleus, then find the value of 500f.

27. An infinite non-conducting plane with uniform charge density σ is kept parallel to yz plane and at a distance 'd' from a dipole $\vec{p}$ which itself is located at the origin. An equipotential surface for this system is spherical, centered at origin, having radius R(<d). Find the value of integer n. Given that $R = \left(\frac{p}{n\pi\sigma}\right)^{1/3}$



28. For $x = a^2y^3z^{1/2}$, where 'a' is a constant. If percentage error in measurement of 'y' and 'z' are 4% and 12% respectively, then percentage error in x will be

29. The least count of main scale of a screw gauge is 1 mm. Find the minimum numbers of divisions on its circular scale required to measure 10 μm diameter of a wire.

30. Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Find the total number of possible different spectral lines in the emission spectra.

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Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

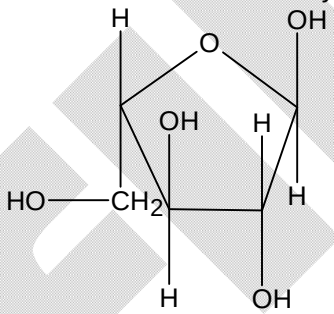
31. The correct relation of O – O bond length is in:
(A) $O_2[AsF_4] < O_2 < KO_2$
(B) $O_2 < O_2[AsF_4] < KO_2$
(C) $KO_2 < O_2[AsF_4] < O_2$
(D) $O_2[AsF_4] < KO_2 < O_2$

32. The zone refining of metal is based on the principle of:
(A) Greater mobility of pure metal then that of impurity.
(B) Higher melting point of the impurity then that of the pure metal.
(C) Greater noble character of the solid metal then that of the impurity.
(D) Greater mobility of the impurity in the molten state then in the solid.

33. XeF_6 dissolve in anhydrous HF to give a good conducting solution which contains:
(A) H^+ and XeF_7^- ion
(B) HF_2^- and XeF_5^+ ions
(C) $HXeF_6^+$ and F^- ions
(D) None of these

34. If pK_b for fluoride ion at 25°C is 10.83, the ionization constant of hydrofluoric acid in water at this temperature is
(A) 1.73×10^{-5}
(B) 3.52×10^{-3}
(C) 6.75×10^{-4}
(D) 5.36×10^{-2}

35. Which set of terms correctly identifies the carbohydrate shown?



1. Pentose
2. Hexose
3. Aldose
4. Ketose
5. Pyranose
6. Furanose

(A) 1, 3 and 6
(B) 1, 3 and 5
(C) 2, 3 and 5
(D) 2, 3 and 6

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36. Which precipitate is white coloured?

(A) $(NH_4)_3PO_4.12MoO_3$ (B) CdS

(C) SnS (D) $Mg(NH_4)PO_4$

37. The standard reduction potential for Cu^{2+}/Cu is +0.34 V. Calculate reduction potential at pH = 14 for the above complex. Given : K_{sp} of $Cu(OH)_2$ is 1×10^{-19}.

(A) –0.2214 V (B) –0.6642 V

(C) –0.4122 V (D) –0.1224 V

38. Select the incorrect option for the following sequence of reaction:

$H-C\equiv C-H \xrightarrow{\text{1. Red hot Fe tube}} P \xrightarrow[\text{2. } , AlCl_3]{\text{3. Zn–Hg/HCl}} Q$

(A) The compound 'P' does not give, 2,4-DNP test.

(B) The compound 'Q' gives effervescence of CO_2 with $NaHCO_3$.

(C) The conversion of P to Q is a type of electrophilic substitution reaction.

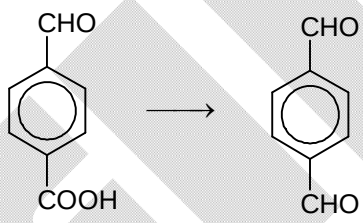
(D) The compound P is soluble in NaOH.

39. Both $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ are diamagnetic. The hybridisation of Nickel in the compound are respectively

(A) sp^3, sp^3 (B) sp^3, dsp^2

(C) dsp^2, sp^3 (D) dsp^2, dsp^2

40. Identify the set of reagents/reaction condition for the following transformations



(A) (i) Conc. HI (ii) red P

(B) (i) PCl_5 (ii) H_2 / Ni

(C) (i) $HO-CH_2-CH_2-OH$ / dry HCl (ii) $SOCl_2$ (iii) H_2 / $Pd-BaSO_4$ (iv) H_3O^+

(D) (i) PCl_5 (ii) $NaBH_4$ / EtOH

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41. For a first order reaction

$X(g) \longrightarrow 2Y(g) + Z(g)$

Initially 3 moles of gas X (g) was taken in a 10 L rigid container at 300 K. After 10 second the reaction was stopped forcefully and made a pinhole in the container at the same temperature. The molar ratio of X(g) and Y(g) coming out initially is 1 : 2. The incorrect statement in the following is:

Given molar mass of gas X, Y and Z are 16, 4 and 8 g/mole respectively, and $\ell n2 = 0.7$, $\ell n3 = 1.1$, $R = 0.08$ atm L^{-1} mol^{-1} K^{-1}

(A) The total pressure inside the container when reaction was just stopped in 12 atm.

(B) Rate constant of the reaction is 4×10^{-2} sec^{-1}

(C) Moles of Z just after reaction was stopped forcefully is 1

(D) The rate of formation of Y after 10 second is 0.4 mole L^{-1} sec^{-1}

42. $NH_3(g) + 3Cl_2(g) \rightleftharpoons NCl_3(g) + 3HCl(g)$; $-\Delta H_1$

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; ΔH_2

$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$; ΔH_3

The heat of formation of $NCl_3(g)$ in terms of ΔH_1, ΔH_2 and ΔH_3 is:

(A) $\Delta H_f = -\Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$

(B) $\Delta H_f = +\Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$

(C) $\Delta H_f = \Delta H_1 - \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$

(D) None of the above

43. Consider the following complex

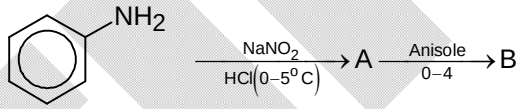
$A = [FeCl_6]^{4-}$, $B = [Fe(H_2O)_6]^{2+}$, $C = [Fe(H_2O)_6]^{3+}$

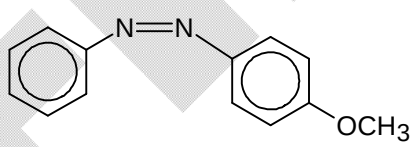
The correct order of decreasing wavelength of light absorbed for the following complex is?

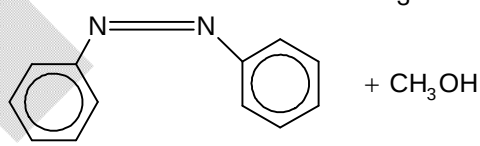
(A) C > B > A (B) A > C > B

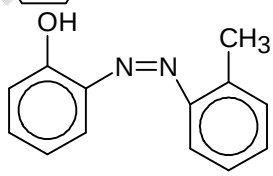
(C) B > A > C (D) A > B > C

44. Identify the coloured compound (B) in the following reaction sequence:

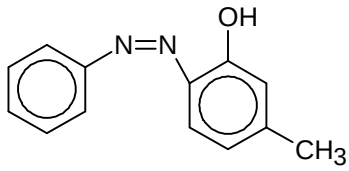


(A) 

(B) 

(C) 

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(D) 

45. For a complex reaction

$A \longrightarrow$ Product

$E_{a_1} = 180$ kJ / mole, $E_{a_2} = 80$ kJ / mole, $E_{a_3} = 50$ kJ / mole

Overall rate constant k is related to individual rate constant by the equation $k = \left(\frac{k_1 k_2}{k_3}\right)^{2/3}$.

Activation energy (kJ/mole) for the overall reaction is

(A) 100 (B) 3.44

(C) 150 (D) 140

46. $A + O_2 \xrightarrow{\text{Exothermic}} \underset{\text{Flower of phosphorous}}{B} + H_2O \longrightarrow C \xrightarrow{-H_2O} \text{Oxy acids} \xrightarrow{-H_2O} D$

Then A, B, C and are respectively?

(A) $B_2H_6, P_4, H_3PO_4, P_2O_5$ (B) P_4, P_4O_{10}, H_3PO_4, P_2O_5

(C) PCl_5, H_3PO_4, HPO_3, P_4O_{10} (D) P_2O_5, P_4O_{10}, H_3PO_4, P_4

47. The half-life of a first order reaction is 120 minute. How long will it take to consume 93.75% of the reactant

(A) 480 min (B) 480 sec

(C) 240 min (D) 240 sec

48. The blue compound formed in the positive test for nitrogen with Lassaigne solution of an organic compound.

(A) $Na_4\left[Fe(CN)_5(NOS)\right]$ (B) $Na_3\left[Fe(CN)_6\right]$

(C) $Fe(CN)_3$ (D) $Fe_4\left[Fe(CN)_6\right].xH_2O$

49. If the 'x is ionization energy of hydrogen, then the energy required for excitation of Li^{2+} electron from 2nd excited state to 5th excited state is

(A) $\frac{3x}{4}$ (B) $\frac{4}{3x}$

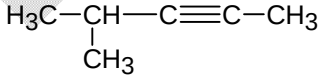
(C) $\frac{x}{12}$ (D) $\frac{12}{x}$

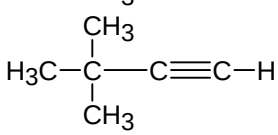
50. $P \xrightarrow[\text{(ii) } NaBH_4/\text{ethanol}]{\text{(i) dil.}H_2SO_4/HgSO_4} Q \xrightarrow[\text{(ii) } O_3,\ Zn/H_2O]{\text{(i) } H^+/\Delta} 2CH_3 - \overset{\overset{O}{\|}}{C} - CH_3$

'P' is

(A) $CH_3 - CH_2 - CH_2 - CH_2 - C \equiv CH$

(B) $CH_3 - CH_2 - C \equiv C - CH_2 - CH_3$

(C) 

(D) 

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SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. 20 cc of hydrocarbon were exploded with excess of oxygen. After explosion and cooling a contraction of 20 cc was noted, and addition of KOH another contraction of 40 cc was noted. If molecular formula of hydrocarbon is C_xH_y then what is the value x + y?

52. If the minimum uncertainty in de-Broglie wavelength of a moving electron accelerate by potential difference of 6 volt. and whose uncertainty in position is $\frac{7}{22}$nm is nearly '0.125x' $\overset{o}{A}$ then find the value of x.

53. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of the solution at 298 K is 2.0 g cm^{-3}, the ratio of the molecular weight of the solute and solvent is

54. If degeneracy of 2nd excited state in H-atom and He-atom is 'x' and 'y' respectively, then the value of 'xy' is

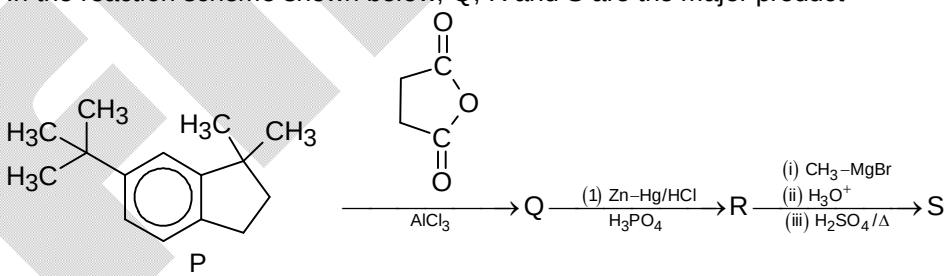
55. $xC_2H_2 \xrightarrow[\substack{\text{ether}\\ 60^oC\\ 10-20\ \text{atm}}]{Ni(CN)_2}$ Product. Find x?

56. On plotting $\frac{\pi_c^2}{\pi^\infty\left(\pi^\infty-\pi^c\right)}$ vs $\frac{1}{c}$

of chloro acetic acid a straight line with slope equal to 0.00014 is obtained. [π^c, π^∞ are molar conductivity of the acid at concentration, 'c' and it infinite dilution, respectively]. If the value of its dissociation constant is $y\times10^{-5}$, then find the value of y?

57. A cube of 1 m edge length is crushed in order to get small cubes of edge length 1 mm. The ratio of surface area (w.r.t. old cube) is 10^z. Then find the value of 'z'.

58. In the reaction scheme shown below, Q, R and S are the major product



Number of degree of unsaturation in product S is…..

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59. (I) $FeSO_4 + KCN(\text{excess}) \longrightarrow (x) \xrightarrow{\text{dil. } H_2So_4} B(g)$

$(x) \xrightarrow{\text{conc. } H_2SO_4} A(g)$

(II) $CaCl_2 + N_2 \xrightarrow{1100^oC} Y + \text{Carbon}$

Find the sum of number of π bonds in A, B and Y.

60. If the number of tetrahedral unit in the structural formula of Borax = p, and if the formula of cyclic cilicate benzyl is $Be_xAl_y(SiO_3)_6$, then the value of p + x + y + z is

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. The number of solutions of the equation $\sin 2x + \cos 2x + \sin x + \cos x + 1 = 0$ in $x \in [0, 2\pi]$ is equal to
(A) 2 (B) 4
(C) 6 (D) 8

62. Let a, b, c d, e be natural numbers in A.P. with common difference 1. If $a + b + c + d + e = f^3$ and $b + c + d = g^2$, (where $f, g \in N$) then the least possible value of 'c' is
(A) 75 (B) 225
(C) 675 (D) 1225

63. Let $\int \frac{3+7x^4}{(x^4+1)x}dx = \ln|g(x)| + c$, then the value of $\frac{g(2)}{g(1)}$ is equal to
(A) 56 (B) 68
(C) 88 (D) 108

64. Tangent PA and PB are drawn from P(h, k) to the parabola $y^2 - 4y - 4x = 0$ such that circumcentre of ΔPAB lies on AB, then the minimum value of $(h^2 + k^2)$ is equal to
(A) 1 (B) 2
(C) 3 (D) 4

65. Number of integral solutions of the equation $\sin^{-1}(|\cos x|) = \cos^{-1}(|\sin x|)$ in $x \in [0, 10\pi]$ is equal to
(A) 12 (B) 20
(C) 30 (D) 32

66. $\int_0^1 \frac{x^5 - x^3}{\log x}dx$ is equal to
(A) $\log\frac{5}{3}$ (B) $\log\frac{3}{5}$
(C) $\log\frac{3}{2}$ (D) $\log\frac{2}{3}$

67. Let the unit vectors $\vec{a}$ and $\vec{b}$ be perpendicular and the unit vector $\vec{c}$ be inclined at an angle θ to both $\vec{a}$ and $\vec{b}$. If $\vec{c} = \alpha\vec{a} + \beta\vec{b} + \gamma(\vec{a} \times \vec{b})$, then
(A) $\alpha = \beta = \cos 2\theta$ (B) $\alpha = \cos\theta, \beta = \sin\theta$
(C) $\gamma^2 = \cos 2\theta$ (D) $\gamma^2 = -\cos 2\theta$

68. The number of unordered pair (m, n) such that factoring of 441000 into two factors, m and n, where $m > 1$, $n > 1$ are relatively prime is
(A) 8 (B) 14
(C) 16 (D) 7

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69. Let $P(\alpha_1,\beta_1,\gamma_1)$ lies on the line $\frac{x-1}{3}=\frac{y-2}{2}=\frac{z-3}{1}$ and $Q(\alpha_2, \beta_2, \gamma_2)$ lies on the line $\frac{x-2}{5}=\frac{y-1}{2}=\frac{z-2}{3}$. If PQ is minimum then the value of $(\alpha_1+\beta_1+\gamma_2)$ is equal to

(A) 1 (B) 5
(C) 3 (D) 4

70. The solution of differential equation $\frac{x+y\frac{dy}{dx}}{y-x\frac{dy}{dx}}=\frac{x\cos^2\left(x^2+y^2\right)}{y^3}$ is

(A) $\tan\left(x^2+y^2\right)=\frac{x^2}{y^2}+c$ (B) $\cot\left(x^2+y^2\right)=\frac{x^2}{y^2}+c$

(C) $\tan\left(x^2+y^2\right)=\frac{y^2}{x^2}+c$ (D) $\cot\left(x^2+y^2\right)=\frac{y^2}{x^2}+c$

71. Let $A=\begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}$ and $C = A^{20}$, then the sum of the elements of C is

(A) 211 (B) 251
(C) 253 (D) 213

72. Let $S_n=\sum_{r=0}^{n} r^2\cdot\left({}^nC_r\right)^2$, then

(A) $S_{20} = 20^{39}C_{20} + 380^{40}C_{20}$ (B) $S_{20} = 19^{40}C_{19} + 20^{39}C_{20}$
(C) $S_{30} = 30^{59}C_{29} + 870^{58}C_{30}$ (D) none of these

73. Let $P(\alpha, \beta)$ be the orthocentre of the triangle with vertices A(–3, 4), $B\left(\sqrt{13},2\sqrt{3}\right)$ and $C\left(2\sqrt{5},\sqrt{5}\right)$, then the value of $[\alpha]$ is equal to (where [.] is G.I.F.)

(A) 4 (B) 5
(C) 6 (D) 7

74. Number of integral values of 'k' in [0, 10] such that $\cos(x)+\cos\left(2\sqrt{k}x\right)=2$ has exactly one real solution is/are

(A) 0 (B) 3
(C) 7 (D) 10

75. Let $P(\alpha, \beta)$ be a fixed point inside the parabola $y^2 + 4x = 0$ with focus S and Q be a point on the parabola such that minimum value of PQ + QS is 4, then $|\alpha|$ is equal to

(A) 1 (B) 2
(C) 3 (D) 4

76. Let the value of $\prod_{r=1}^{45}\sin\left(2r-1\right)^0=2^{-\alpha}$, then the value of $[\alpha]$ is equal to (where [.] is G.I.F.)

(A) 40 (B) 42
(C) 44 (D) 46

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77. Which is a statement ?
(A) x = 1 (B) x + y = 10
(C) 2 is not an integer (D) none of these

78. If the vectors $\vec{a},\vec{b},\vec{c}$ satisfy the condition $|\vec{a}-\vec{c}|=|\vec{b}-\vec{c}|$, then the value of $(\vec{a}-\vec{b})\cdot\left(\frac{\vec{a}+\vec{b}}{2}-\vec{c}\right)$ is equal to
(A) –1 (B) 0
(C) 1 (D) 2

79. Number of real solutions of the equation $\log_2\left(x-5^{\log_{10}x}\right)=\log_{10}x$ is/are
(A) 0 (B) 1
(C) 2 (D) none of these

80. If $\int\frac{x^6+4x^3}{\left(x^4+x^3+1\right)^2}dx=\frac{f(x)}{x^4+x^3+1}+\lambda x+c$, where f(x) is a polynomial of degree less than four, then the value of f''(–1) + λ is equal to
(A) 6 (B) 12
(C) 18 (D) 24

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Number of integral values of 'k' such that the equation $(2x^3 – 9x^2 + 12x + 1)^2 – 2k(2x^3 – 9x^2 + 12x + 1) + (k^2 – 1) = 0$ has 6 real roots is

82. If $\sum_{k=1}^{n}\frac{2^k\left(k^2+2k+2\right)}{(k+1)(k+2)}=2^{15}-2^{11}-1$, then 'n' is equal to

83. If the area bounded by the real curve $5x^2 + 5y^2 + \lambda xy – 4x – 4y + 4 = 0$ and the line $y= mx + 1 – m$ is independent of 'm', then the value of $|\lambda|$ is equal to

84. Let $\alpha_k = 3i + e^{i\frac{2k\pi}{13}}$, $k \in N$, $k \le 100$, then number of values of 'k' such that $|\alpha_k + 3 – 3i|$ is minimum is/are

85. Let $(x^2 + 2x + 2)^{20} = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{40}x^{40}$, then the value of $(20a_0 – a_1 + a_{39})$ is equal to

86. If the value of $\int_0^{\pi/2}\sqrt{\frac{\cos x-\cos^3 x}{1-\cos^3 x}}dx$ is equal to $\frac{k\pi}{6}$, then the value of 'k' is equal to

87. The mean monthly salary paid to all employees of a company is Rs. 8400. The mean monthly salary paid to male and female employees are Rs. 8000 and Rs. 9000 respectively. If ratio of number of male employees to number of female employees is $\frac{k_1}{k_2}$, (k_1 and k_2 are co-prime), then the value of '$k_1 – k_2$' is equal to

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88. A natural number is selected at random from first 100 natural number, if the probability that the selected number satisfies the relation $x + \frac{16}{x} > 10$ is 'P', then the value of $\frac{100P}{3}$ is

89. The length of the sides of a triangle are b + 1, 7 – b and 4b – 2. The number of values of 'b' for which the triangle is isosceles is

90. Let A(2, 0) and B(–2, 0) be two points such that a variable point P satisfies the relation PA – PB = $2\sqrt{2}$. The number of integral values of α for which exactly two tangents are drawn from the point $Q(\alpha^2, \alpha + 1)$ to the locus of 'P' is

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